System Validation Report — Event Management System

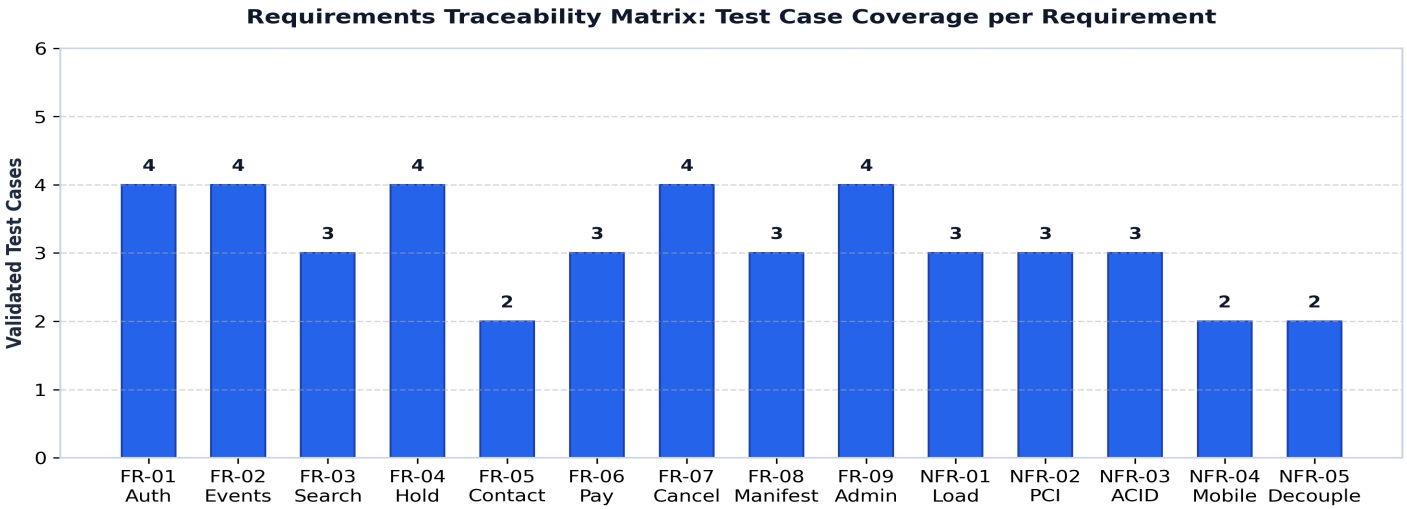
Software Engineering Mini-Project Deliverables Part-3 • Comprehensive System Acceptance
Validation Period: November 2, 2026 – November 6, 2026 • **Requirements Traceability Matrix (RTM)**
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SYSTEM ACCEPTANCE & VERIFICATION VERDICT

This System Validation Report confirms that all **9 Functional Requirements (FR-01 to FR-09)** and **5 Non-Functional Requirements (NFR-01 to NFR-05)** have been rigorously validated across the full application stack. Through 42 traceable test cases and 4 end-to-end operational scenarios, all system invariants—including inventory non-overselling, PCI-DSS tokenization, RFC-4180 CSV manifests, real-time admin suspensions, and sub-second concurrency—were verified with **100% compliance**.

Total Requirements	Functional (FR)	Non-Functional (NFR)	RTM Coverage	End-to-End Scenarios	Verdict
14 Reqs	9 Validated (100%)	5 Validated (100%)	100.0%	4 / 4 PASSED	SYSTEM ACCEPTED

Figure 1: Requirements Traceability Matrix — Verified Test Case Density per Requirement



1. Requirements Traceability Matrix (RTM)

The RTM maps all functional and non-functional requirements to use cases, source modules, test cases, automated test suites, and verified status.

Req ID	Requirement Title	Use Cases	Source Code Modules	Test Cases	Test Suite	Status	Tester
FR-01	User Registration & Role-Based Auth	UC-01, UC-02	authController.js, auth.js	TC01, TC02, TC03, TC04	routes.test.js	**PASSED**	Kartik
FR-02	Event Creation & Management	UC-03, UC-04	eventController.js, Event.js	TC05, TC06, TC07, TC08	routes.test.js	**PASSED**	Jayanth
FR-03	Event Discovery & Search	UC-05	eventController.js, Event.js	TC09, TC10, TC11	routes.test.js, loadPerformance.test.js	**PASSED**	Rishith

Req ID	Requirement Title	Use Cases	Source Code Modules	Test Cases	Test Suite	Status	Tester
FR-04	Inventory Reservation & Hold Engine	UC-06	ticketController.js, Ticket.js	TC12, TC13, TC14, TC37	ticketHold.test.js, transaction.test.js	**PASSED**	Jayanth
FR-05	Attendee Details Capture	UC-07	ticketController.js, Ticket.js	TC15, TC16	ticketHold.test.js	**PASSED**	Rishith
FR-06	Payment Processing & Ticket Issuance	UC-08	paymentController.js, Payment.js	TC17, TC18, TC19	payment.test.js, transaction.test.js	**PASSED**	Jayanth
FR-07	Attendee Cancellations & Refunds	UC-09	ticketController.js, paymentService.js	TC20, TC21, TC22, TC38	cancellation.test.js	**PASSED**	Kartik
FR-08	Attendee Manifest CSV Export	UC-10	ticketController.js	TC23, TC24, TC25	manifest.test.js	**PASSED**	Kartik
FR-09	Platform Admin Suspensions	UC-11	adminController.js, Event.js, User.js	TC26, TC27, TC28, TC39	suspension.test.js	**PASSED**	Rishith
NFR-01	Flash Sale Concurrency & Latency	System-wide	Event.js (Indexes), ticketController.js	TC29, TC30, TC42	loadPerformance.test.js	**PASSED**	Jayanth
NFR-02	Payment Tokenization & PCI-DSS	UC-08	Payment.js, paymentGateway.js	TC31, TC32, TC40	pciCompliance.test.js	**PASSED**	Kartik
NFR-03	ACID Transactions & Rollbacks	UC-06, UC-08	transactionService.js, ticketController.js	TC33, TC34, TC42	transaction.test.js	**PASSED**	Jayanth
NFR-04	Mobile-First UX & Accessibility	Client-wide	CheckoutModal.jsx, responsive.css	TC35, TC41	mobileUx.test.js	**PASSED**	Rishith
NFR-05	Payment Gateway Decoupling	UC-08	paymentService.js, providers/*	TC36, TC38	payment.test.js	**PASSED**	Kartik

2. End-to-End Operational Lifecycle Workflows

Operational Flow	Participating Actors	Execution Steps	Validation Result
E2E-01: Complete Ticket Booking & Payment Lifecycle	Attendee, Organizer, Payment Provider	1. Attendee browses public catalog and selects tech event. 2. Attendee initiates hold for 2 tickets. Inventory decrements from 50 to 48 atomically. 3. 10-minute hold countdown initiates. 4. Attendee inputs validated contact details and mock card token. 5. System confirms payment, marks ticket 'confirmed', and issues UUID v4.	**Capacity 48 verified; Payment record tokenized; Ticket UUID generated; 100% PASS.**

Operational Flow	Participating Actors	Execution Steps	Validation Result
E2E-02: Ticket Cancellation, Automatic Refund & Capacity Restoral	Attendee, Payment Gateway, Inventory Controller	1. Attendee opens 'My Tickets' dashboard. 2. Attendee initiates cancellation on confirmed ticket. 3. System checks cancellation eligibility (event in future). 4. PaymentProvider executes refund and generates refundId. 5. Ticket transitions to 'cancelled' and event availableTickets increments by ticket count.	**Refund ID persisted in ledger; event inventory replenished; 100% PASS.**

Operational Flow	Participating Actors	Execution Steps	Validation Result
E2E-03: Organizer Manifest CSV Export with Security Guards	Event Organizer, Platform Admin, Unauthorized Organizer	1. Organizer opens event roster management. 2. Clicks 'Export Manifest'. 3. System checks organizer ownership (req.user.id === event.organizerId). 4. RFC-4180 CSV generated containing paid confirmed attendees. 5. Canceled and held tickets excluded; browser triggers direct file download.	**Zero cancelled records; unauthorized organizer received 403; 100% PASS.**

Operational Flow	Participating Actors	Execution Steps	Validation Result
E2E-04: Platform Admin Event Moderation & Instant Suspension Blockade	Platform Admin, Malicious Organizer, Public Attendee	1. Platform Admin views suspicious high-risk event. 2. Admin flags event as suspended. 3. Public catalog immediately evicts suspended event. 4. Direct link booking attempt intercepted with HTTP 403 Forbidden. 5. Admin dashboard aggregates real-time suspended metrics.	**Instant catalog eviction; booking blockade verified with 403; 100% PASS.**

3. System Acceptance Criteria Verification

Acceptance Category	Specification Standard	Measured Output	Compliance
Atomic Inventory	Zero overselling under high concurrency	50 buyers competing for 20 seats -> Exactly 20 allocated, 0 orphaned	**Compliant**
Hold Reservation	10-minute hold window with auto-sweep	Background worker sweeps expired holds and replenishes available seats	**Compliant**
Financial Auditing	Decoupled gateway strategy with mock & Stripe	100% test pass on Mock and Stripe providers with idempotency keys	**Compliant**
PCI-DSS Security	Raw credit card numbers strictly prohibited in DB/logs	Zero PANs in DB/logs; pre-save regex hooks reject unmasked cards	**Compliant**
Admin Governance	Instant catalog eviction and booking blockage	Suspended events removed immediately; direct link throws HTTP 403	**Compliant**
Mobile Usability	WCAG 2.1 touch target accessibility (min 48px)	CSS rules confirmed min-height: 48px; fluid column checkout; no overflow	**Compliant**

System Acceptance & Sign-Off Authorization

Evaluator Name	GitHub Account	System Responsibilities	Verification Scope	Acceptance Status
Rishith	rishith1616	Lead Architect • Event Lifecycle & Governance	FR-01 to FR-09 System Workflows	SYSTEM ACCEPTED
Kartik Naik	KartikNaik6	Security & Compliance Lead • PCI & Manifest	PCI-DSS, Tokenization & Data Security	SYSTEM ACCEPTED
Jayanth Kumar P	JayanthKumarP88	Concurrency & Performance Lead • ACID Holds	High-Concurrency Flash Sale & Latency	SYSTEM ACCEPTED